Reg. No. | | | | | | | | | | | [14AE65]

SIXTH SEMESTER B.TECH EXAMINATION, JUNE–2019

FRACTURE MECHANICS

Time:3 Hours **Maximum Marks:70**

Instructions: Answer any five full questions selecting at least two full questions from each part.

PART - A

1. a. Describe the intergranular and transgranular fracture **05**
   b. Mention the techniques of failure analysis by conventional approach **04**
   c. Explain the modern design approach **05**

2. a. Explain the Griffith theory of fracture **05**
   b. Obtain the energy release rate equation for a DCB specimen loaded with a constant load **04**
   c. Explain the concepts of G & R **05**

3. a. Derive the mode I stress intensity factor crack in an infinite plate where the normal to the crack plane is oriented at an angle β with the stress axis. **04**
   b. Explain the strip yield model of the crack tip plasticity **04**
   c. A large plate of 5mm thickness, made of medium carbon steel, σ_{ys}=350MPa with a through the thickness centre crack of 2a=40mm length is subjected to a stress of 150MPa. For mode I loading, determine the effective crack length using Irwin's correction. **06**

4. a. Describe the methodology to estimate CTOD experimentally and obtain an expression for total CTOD. **05**
   b. Prove that the J-contour integral is path independent around any contour. **04**
   c. Obtain the relationship between J & CTOD. **05**

PART - B

5. a. Define Stress range, K_{max}, K_{min}, Stress Intensity Factor range, Stress ratio **05**
   b. Obtain an expression for crack propagation life N_P **05**
   c. The fatigue data for a brass alloy are given as follows, **04**

Stress Amplitude (S) (MPa)	Cycles to Failure (N)
310	$2x10^5$
223	$1x10^6$
191	$3x10^6$
168	$1x10^7$
153	$3x10^7$
143	$1x10^8$
134	$3x10^8$
127	$1x10^9$

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i. Make an S-N plot using the data.

ii. Determine the fatigue strength at $5x10^5$ cycles.

iii. Determine the fatigue life for 200MPa.

6. a. Explain the process zone mechanism for ceramic toughening. **07**

b. Describe transformation toughening **07**

7. a. Describe Fiber and Whisker toughening **07**

b. Explain the principles of fracture safe design **07**

8. Write short notes on

i. Dye-penetrant inspection **05**

ii. Magnetic-particle inspection (magnaflux) **05**

iii. Effect of an underload **04**